# JavaScript – Number-Based Programs

Five beginner-friendly while-loop programs with the purpose, step-by-step logic, code, and sample output. Practise one program at a time.

## Important operators

**%** gives the remainder. JavaScript uses **Math.floor(num / 10)** to remove the last digit for positive whole numbers.

## 1. Reverse a Number

**What this program does:** Reverse the digits of a number. Example: 1234 becomes 4321.

**Logic:**

1. Get the last digit using remainder (%) with 10.
2. Add that digit to the reversed number after shifting the previous digits left.
3. Remove the last digit using floor division (// in Python; Math.floor division in JavaScript).
4. Repeat until the original number becomes 0.

## JavaScript code:

```
let num = Number(prompt("Enter a number:"));
let rev = 0;

while (num > 0) {
    let digit = num % 10;
    rev = rev * 10 + digit;
    num = Math.floor(num / 10);
}

console.log("Reverse is:", rev);
```

## Sample output:

```
Enter a number: 1234\nReverse is: 4321
```

**Practice tip:** Trace the values of `num`, `digit`, and the result variable after each loop iteration.

## 2. Count the Digits

**What this program does:** Count how many digits are present in a number. Example: 458 has 3 digits.

**Logic:**

1. Start the count at 0.
2. Each time through the loop, remove the last digit and increase the count by 1.
3. Stop when the number becomes 0.
4. Special case: 0 itself has 1 digit.

## JavaScript code:

```
let num = Number(prompt("Enter a number:"));
let count = 0;

if (num === 0) {
    count = 1;
} else {
    while (num > 0) {
        count = count + 1;
        num = Math.floor(num / 10);
    }
}

console.log("Number of digits:", count);
```

## Sample output:

```
Enter a number: 458\nNumber of digits: 3
```

**Practice tip:** Trace the values of `num`, `digit`, and the result variable after each loop iteration.

## 3. Sum of Digits

**What this program does:** Add all the digits of a number. Example: 458 gives 4 + 5 + 8 = 17.

**Logic:**

1. Extract the last digit using % 10.
2. Add the extracted digit to sum.
3. Remove the last digit using // 10 in Python or Math.floor(num / 10) in JavaScript.
4. Repeat until the number becomes 0.

## JavaScript code:

```
let num = Number(prompt("Enter a number:"));
let total = 0;

while (num > 0) {
    let digit = num % 10;
    total = total + digit;
    num = Math.floor(num / 10);
}

console.log("Sum of digits:", total);
```

## Sample output:

```
Enter a number: 458\nSum of digits: 17
```

**Practice tip:** Trace the values of `num`, `digit`, and the result variable after each loop iteration.

## 4. Check a Palindrome Number

**What this program does:** Check whether a number reads the same forward and backward. Example: 121 is a palindrome.

**Logic:**

1. Keep the original number safe in a separate variable.
2. Reverse the number using the digit-extraction method.
3. Compare the reversed number with the original number.
4. If they are equal, it is a palindrome; otherwise, it is not.

## JavaScript code:

```
let num = Number(prompt("Enter a number:"));
let original = num;
let rev = 0;

while (num > 0) {
    let digit = num % 10;
    rev = rev * 10 + digit;
    num = Math.floor(num / 10);
}

if (original === rev) {
    console.log("Palindrome");
} else {
    console.log("Not a palindrome");
}
```

## Sample output:

```
Enter a number: 121\nPalindrome
```

**Practice tip:** Trace the values of `num`, `digit`, and the result variable after each loop iteration.

## 5. Product of Digits

**What this program does:** Multiply all the digits of a number. Example: 234 gives 2 × 3 × 4 = 24.

**Logic:**

1. Start product at 1 (not 0, because multiplying by 0 makes the result 0).
2. Extract the last digit using % 10.
3. Multiply product by the extracted digit.
4. Remove the last digit and repeat.

## JavaScript code:

```
let num = Number(prompt("Enter a number:"));
let product = 1;

if (num === 0) {
    product = 0;
} else {
    while (num > 0) {
        let digit = num % 10;
        product = product * digit;
        num = Math.floor(num / 10);
    }
}

console.log("Product of digits:", product);
```

## Sample output:

```
Enter a number: 234\nProduct of digits: 24
```

**Practice tip:** Trace the values of `num`, `digit`, and the result variable after each loop iteration.